

Letter to the Reader

To Whom It May Concern,

Thank you for taking the time to look though my project documentation. Your interest in reading more about my project makes all of the work I put into it worth it. I hope that you find what you are looking for or encounter something you weren't expecting.

Please do not hesitate to contact me if you have any questions or comments.

Sincerely,

A handwritten signature in black ink, appearing to read 'Meagin Arrocha', with a stylized, cursive script.

Meagin Arrocha

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Senior Project History

I started really thinking about what I wanted to do for my project toward the end of Sophomore year. It took a lot of time to figure out what I wanted to do and I went through a lot of ideas before I decided to make a website. Having spent three weeks in Panama with my husband, I saw how the animals were treated there and I wanted to eventually make an animal shelter there. Eventually I needed a website for the company so I started making a website in Ruby on Rails. After doing about a whole semester's worth of work on it, I decided I did not like that idea and it was not proving to be a worthy enough senior project. It was also difficult to create it in Ruby on Rails and using Github because I could never get all of my code to work together. Long story short, I really wanted to do something else. Thus, this project was born.

It began when I was watching a video on YouTube about a chess game being played where both players had their own huge chess board with both sets of pieces on it: theirs and their opponents. When one player would move their chess piece, that corresponding chess piece on the other players board would move. This occurred for each turn and they were playing the game together even though they were sitting apart from each other. It made me think about how cool it would be to play a board game with someone who is not in the same room with you. I thought about how many children, especially those without siblings, have parents who have to travel for long periods of time and don't get to spend as much quality time with them as they might like. Therefore, I wanted to make a game to do that.

The game I chose to base my project on was Snakes and Ladders. It is also known as Shoots and Ladders but instead of shoots there are snakes. Using the same game rules and concepts. I wanted to make a piece that was movable on the child's side so when their parent, for example, would take their turn, the movable piece would move on its own as if they were there playing with them. Another reason I selected this game in particular was because, unlike a game like chess, there is no decision making in this game. All of the moves are based on die rolls which makes it easier for a computer to

control since it is just mathematical equations. However, this project has turned out to be a lot more difficult than I anticipated it being. There are so many little components to factor in and it seems even when you go one step forward you go twelve steps back. With that being said, I have learned a lot from this project; not only about the hardware and software that I used, but about myself. When things get hard I notice that I tend to give up or put it to the side and I never get back to it. This project has taught me to stick through and see it to the end even if the outcome is not ideal.

Hardware/Software

Hardware:

- **Arduino Uno**
 - Provided by Dr. Sessions
 - Bottom layer attached to car
 - Responsible for holding the code and communicating between the Wi-Fi shield and the Motor Shield
- **Arduino WiFi Shield**
 - Provided by Dr. Sessions
 - Middle layer attached to car
 - Responsible for communicating with the router
 - Establishing a connection
 - Receiving packets
- **D1203 Motor Shield**
 - Purchased by me
 - Top layer attached to car
 - Responsible for allowing two DC motors to be driven with the Arduino Uno
 - Allows control of speed and direction of motors
- **Chassis Car**
 - Purchased by me
 - Necessary for task to be completed
 - Responsible for holding equipment on top and for the visual aspect of the projects
- **Router**
 - provided by Dr. Lin

- Responsible for connecting the C++ code to the Arduino Uno

Software:

- **Arduino 1.6.6 IDE**
 - Free to download
 - Responsible for writing the code for the Arduino Uno
 - Provides ability for Uno, Motor and WiFi shields to communicate
 - Written in a C/C++ like language
- **Visual Studio C++ Environment**
 - Community is free to download
 - Responsible for the “game board functions” like the position of the car
 - Takes into consideration the position changes due to hitting a ladder or a snake head
- **Visual Studio Android Developing Environment**
 - Free update for VS called Xamarin allows native Android, iOS, and Windows apps development
 - Responsible for creation of future phone app to play the game with

Arduino Code

/*

WiFi UDP Send and Receive String

This sketch wait an UDP packet on localPort using a WiFi shield.

When a packet is received an Acknowledge packet is sent to the client on port remotePort

Circuit:

* WiFi shield attached

created 30 December 2012

by dlf (Metodo2 srl)

<https://www.arduino.cc/en/Tutorial/WiFiSendReceiveUDPString>

*/

```
#include <SPI.h>
```

```
#include <AFMotor.h>
```

```
#include <WiFi.h>
```

```
#include <WiFiUdp.h>
```

```
char ssid[] = "TP-LINK"; //router network SSID name
```

```
char pass[] = "password"; //router password
```

```
int status = WL_IDLE_STATUS;
```

```
unsigned int localPort = 2390; // local port to listen on
```

```
int connectPin = 7;
```

```
AF_DCMotor M1Motor(1);
```

```
AF_DCMotor M2Motor(2);
```

```
WiFiUDP Udp;
```

```
char packetBuffer[255]; //buffer to hold incoming packet
```

```
void setup(){
```

```
  // initialize serial:
```

```
  Serial.begin(9600);
```

```
  // attempt to connect using WPA2 encryption:
```

```
  Serial.println("Attempting to connect to WPA network...");
```

```
  status = WiFi.begin(ssid, pass);
```

```

// if you're not connected, stop here:
if ( status != WL_CONNECTED) {
  Serial.println("Couldn't get a wifi connection");
  while(true);
}
// if you are connected, print out info about the connection:
else {
  Serial.println("Connected to network");
  printWifiStatus(); //prints IP address
  Udp.begin(localPort);
}

```

```

M1Motor.setSpeed(200);
M1Motor.run(RELEASE);
M2Motor.setSpeed(200);
M2Motor.run(RELEASE);

```

```

}

```

```

void loop(){
//if there is data to available, read a packet
int packetSize = Udp.parsePacket();
if (packetSize){
  Serial.print(F("Received packet of size "));
  Serial.print(packetSize);
  Serial.print(F("From"));
  Serial.print(F(", port"));
  Serial.println(Udp.remotePort());

```

```

//read the packet into packetBuffer
int len = Udp.read(packetBuffer, 255);
if (len > 0) packetBuffer[len] = 0;
Serial.print(F("Contents: ["));
Serial.print(packetBuffer);
Serial.println(F("]"));
char c = packetBuffer[0];

```

```

//use first character to decide movement
if (c == 'f'){
  M1Motor.run(BACKWARD);
  M2Motor.run(BACKWARD);
  Serial.println("go forward");
}
else if(c == 'b'){
  M1Motor.run(FORWARD);
  M2Motor.run(FORWARD);

```



```

        Serial.println("backwards");
    }
    else if(c == 'l'){
        M1Motor.run(RELEASE);
        M2Motor.run(BACKWARD);
        Serial.println("left");
    }
    else if(c == 'r'){
        M1Motor.run(BACKWARD);
        M2Motor.run(RELEASE);
        Serial.println("right");
    }
    else if (c == 's'){
        M1Motor.run(RELEASE);
        M2Motor.run(RELEASE);
        Serial.println("stop");
    }
    else{
        Serial.println("err: did not understand command: [%s]\n");
    }
}

}

void printWifiStatus(){
    //print Wifi shield's IP address:
    IPAddress ip = WiFi.localIP();
    Serial.print("IP Address: ");
    Serial.println(ip);
}

```

C++ Code

```
#include <stdlib.h>

#include <iostream>
#include <time.h>

using namespace std;

int board[5][6];

void createBoard() {
    for (int i = 0; i < 5; i++) {
        for (int j = 0; j < 6; j++) {
            board[i][j] = 0;
            // cout << board[i][j];
        }
        //cout << endl;
    }
}

void printBoard() {
    for (int i = 0; i < 5; i++) {
        for (int j = 0; j < 6; j++){
            cout << board[i][j];
        }
        cout << endl;
    }
}

void main() {
    createBoard(); //make 1-30 board

    srand(time(NULL)); //get time for random number
    int roll; //random number between 1 and 6 for dice roll
    //cout << roll << endl;

    //correspond each position to a number
    //ex: #25 is [0][0]

    int position = 1; // position of car; starts at one spot
    // position is determined by current position plus roll
    //unless in case of certain circumstance (ie. snake or ladder)

    board[4][0] = 1; //starting position

    while (position < 30) {
        printBoard();
        cout << endl;
        cout << endl;

        createBoard();
        cout << endl;
        cout << endl;

        cout << "position = " << position << endl;
```

```

roll = rand() % 6 + 1;
position += roll;
cout << "roll = " << roll << endl;
cout << endl;
//0 on board = empty spot
//1 on board = current position
switch (position) {
case 1: //start
    board[4][0] = 1;
    break;
case 2:
    board[4][1] = 1;
    break;
case 3: //ladder, go to position 22
    //board[4][2] = 1;
    board[1][2] = 1;
    position = 22;
    break;
case 4:
    board[4][3] = 1;
    break;
case 5: //ladder, go to position 8
    //board[4][4] = 1;
    board[3][4] = 1;
    position = 8;
    break;
case 6:
    board[4][5] = 1;
    break;
case 7:
    board[3][5] = 1;
    break;
case 8:
    board[3][4] = 1;
    break;
case 9:
    board[3][3] = 1;
    break;
case 10:
    board[3][2] = 1;
    break;
case 11: //ladder, go to position 26
    //board[3][1] = 1;
    board[0][1] = 1;
    position = 26;
    break;
case 12:
    board[3][0] = 1;
    break;
case 13:
    board[2][0] = 1;
    break;
case 14:
    board[2][1] = 1;
    break;
case 15:
    board[2][2] = 1;
    break;
case 16:
    board[2][3] = 1;

```

```

        break;
    case 17://snake, go to position 4
        //board[2][4] = 1;
        board[4][3] = 1;
        position = 4;
        break;
    case 18:
        board[2][5] = 1;
        break;
    case 19://snake, go to position 7
        //board[1][5] = 1;
        board[3][5] = 1;
        position = 7;
        break;
    case 20: //ladder, go to position 29
        //board[1][4] = 1;
        board[0][4] = 1;
        position = 29;
        break;
    case 21://snake, go to position 9
        //board[1][3] = 1;
        board[3][3] = 1;
        position = 9;
        break;
    case 22:
        board[1][2] = 1;
        break;
    case 23:
        board[1][1] = 1;
        break;
    case 24:
        board[1][0] = 1;
        break;
    case 25:
        board[0][0] = 1;
        break;
    case 26:
        board[0][1] = 1;
        break;
    case 27://snake, go to position 1
        //board[0][2] = 1;
        board[4][0] = 1;
        position = 1;
        break;
    case 28:
        board[0][3] = 1;
        break;
    case 29:
        board[0][4] = 1;
        break;
    }
    system("pause");
}

board[0][5] = 1;
printBoard();

cout << "position = " << position << endl;
cout << "you win!" << endl;
}

```

Event Log

Saturday, August 22, 2015

Decided on Snakes and Ladders Project. Spent 2 hours researching other projects using car models.

Tuesday, September 1, 2015

Met up with Dr. Sessions. Spent 1 hour talking about new project and what direction to go in. She gave me advice on where I should start and also gave me some Arduino parts to use.

Saturday, September 12, 2015

Spent 2 hours downloading the Arduino IDE and using the Arduino Uno and circuit board to get a feel for how the environment worked.

Sunday, October 4, 2015

Ordered Chassis Car and waited on its arrival. Spent about an hour starting my PowerPoint presentation.

Friday, October 23, 2015

Car arrived and I realized the instructions were in Chinese. Spent 3 hours looking around on the internet at videos putting the car together and trying to find the instructions in English.

Wednesday, November 11, 2015

Met with Dr. Lin who showed me Google translate. This would translate the Chinese instructions into English for me when I took pictures of them. Spent 1 hour trying to decipher the broken English it was translated into and decided I could figure it out between the internet and the pictures on the

instructions.

Thursday, November 26, 2015

Met with Dr. Sessions again for about an hour to discuss what I needed to do over Christmas break. She also gave me some more Arduino stuff in case I needed it over the break.

The Month of December 2015 into January 2016

Spent about 2 hours putting the actual car together. Spent another 2 and a half hours trying to figure out how to make the car move and realized I needed a motor shield. I then ordered that part and waited for it to come in.

Saturday, January 23, 2016

Motor Shield Came in. Spent half an hour checking to make sure the 4 DC motors I was given worked. Spent half an hour hooking up all four motors. Spent 2 hours trying to figure out why they all wouldn't work at the same time.

Monday, January 25, 2016

Met with Dr. Sessions for about an hour. We took off two of the motors and got two to run in the back and put a wheel on the front so the car could turn left and right. She also gave me a Wi-Fi shield at this time.

Sunday, January 31, 2016

Spent 3 hours hooking up the Wi-Fi shield, looking up tutorials and guides on it, and trying to get my car to communicate with everything else I had.

Tuesday, February 16, 2016

Spent 3 hours working on an Android app prototype that could have been potentially used for this project. I didn't get very far with it though having to learn how to use Xamarin in Visual Studio.

Thursday, February 25, 2016

Spent an hour writing C++ code to show the position of the car and to take into account the position changes when running into a ladder or a snake. This code starts the car in the first position, shows the position of the car throughout the game, and lets the user know when they have one; showing the car in the final position.

Sunday, March 6, 2016

Came back to working with the Wi-Fi shield and after chatting with classmate Joshua, spent about 2 hours trying connecting to a router and using NetCat to talk the Arduino.

Saturday, March 19, 2016

Spent 2 hours trying to update the firmware on my Wi-Fi Shield and trying again to communicate from my computer to the router and then to the Wi-Fi shield and Arduino.

Saturday, March 26, 2016

Updated my PowerPoint that I started at the end of last year.

Monday, April 11, 2016

Met up with fellow student, Joshua, for about an hour who helped me try to use netcat to communicate with my Arduino.

Tuesday, April 19, 2016

Met up with Dr. Sessions for 6 hours who helped me try to get my C++ code to work with my Arduino.

Thursday, April 21, 2016

Spent an hour double checking my documentation was ready and proofreading it. Also I made sure I had everything I needed for the presentation the next day.

Friday, April 22, 2016

Presented project to Computer Science Faculty and peers.

Total approximate hours spent: 40 hours

Total approximate cost: \$40 (not including parts loaned to me)

Future Tasks & Tentative Add-ons

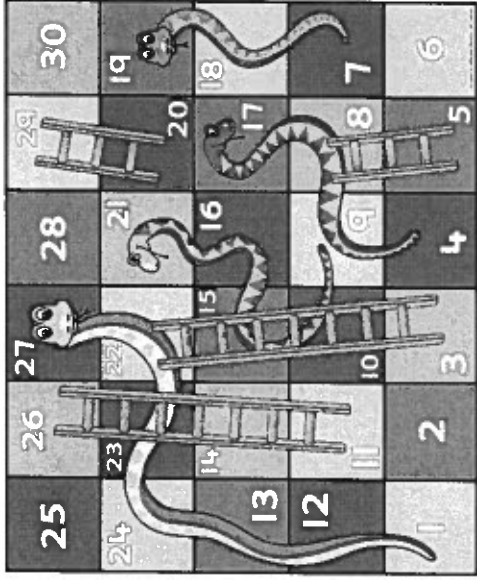
In the future, these are the features that need to be developed to make the game more complete:

- Creation of physical board game
- Completion of phone application to play the game on
- A better choice of colors for the game. Maybe hiring a designer to design the game board and coordinate colors
- Possibility to connect from anywhere in the world and play with someone from anywhere in the world
- A rule/instruction booklet

I would like to see these add-on features that would make the experience better for the users:

- More than two player compatibility
- A smaller scaled game instead of the big car and game board
- Ability to play as a normal board game as well as it is now to appeal to a wider audience
- A different movable piece type
 - i.e. something besides a car that goes better with the theme of snakes and ladders
- A few different ways to play the game in the rule book to keep it interesting

Snakes and Ladders



What is the Idea?

- Snakes and Ladders game
 - Board game
 - Movable piece for remote player
 - Another “normal” piece for local player

Why Make This?

- Senior Project
- YouTube Video
 - Chess game
- No decision making process
 - Just math

Who is it for?

- Audience : Children
 - Only/Lonely children
 - Working Parent(s)
- Two Players
 - Both players
 - Tentative for more players



Hardware/Software

- Hardware:
 - Arduino Uno
 - Arduino WiFi Shield
 - D1203 Motor Shield
 - Chassis Type Car
 - Router
- Software:
 - Arduino 1.6.6 IDE
 - Visual Studio C++ Environment
 - Visual Studio Android Developing Environment

Challenges

- Coming up with an idea
- Foreign Hardware
 - Never using before
- Construction of board
 - I can't draw
- No knowledge of wiring or circuits
- Instructions in Chinese
- Communicating over a network



Future Tasks/Necessities

- Completion of physical board game
- Completion of phone application
- Getting a designer to design the game board and coordinate colors
- Connecting from anywhere in the world to play
- A rule/instruction booklet

Future Add-ons

- Phone app
- More than two players
- Smaller scaled game
- Ability to play as normal board game
 - Appeal to wider range of consumers
- Different type movable piece
 - Something instead of a car to go with theme
- Few different ways to play the game to keep it interesting

Documentation

- To read more in depth about the project, please see the documentation binder I brought with me today.

ZK-4 小车底盘说明图

非常感谢您购买 ZK-4 智能小车，这款小车是为机器人入门专门打造的高性价小车底盘，小车的电机能支持 3V~12V 的供电系统，并且带有测速口，舵机口，寻迹，避障固定位等；方便您的拓展试验。收取产品后应第一时间参照配置清单检查元件是否齐全，若发生缺少元器件情况请联系经销商。请您在安装的时候参照说明书安装，以便快速安装使用。



M3*30 铜柱 4 个



M3*8 螺丝 8 个



M3*30 螺丝 8 个



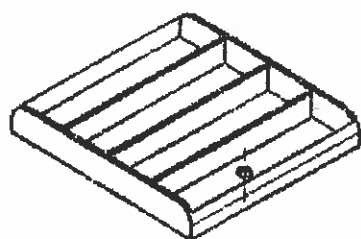
M3 螺母 8 个



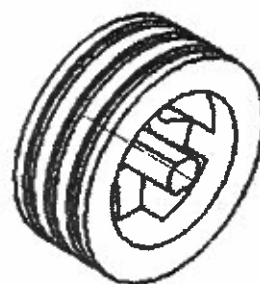
码盘 4 个



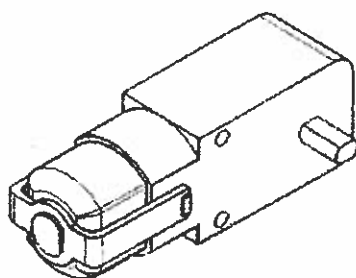
T 型支架 8 个



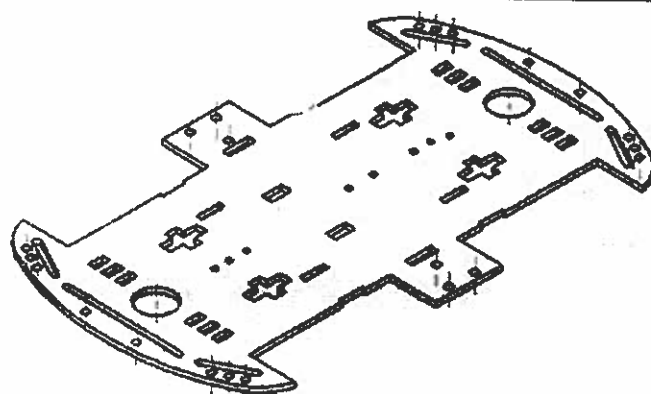
电池盒 1 个



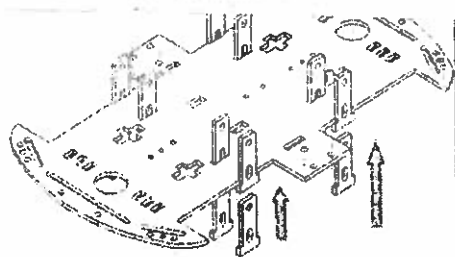
轮胎 4 个



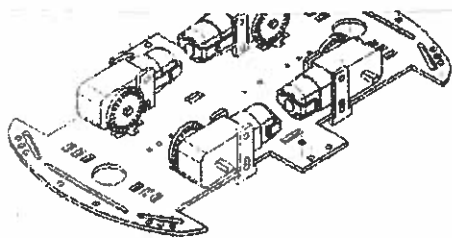
马达 4 个



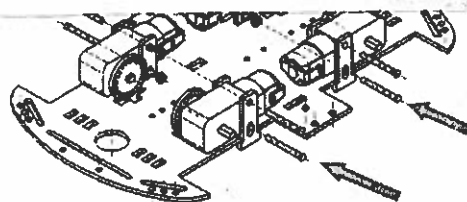
底盘 2 个



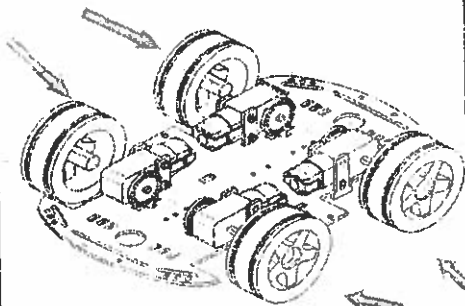
1、先把小车底盘、紧固件、码盘的黄色保护纸撕掉，然后把紧固件插入小车底盘。



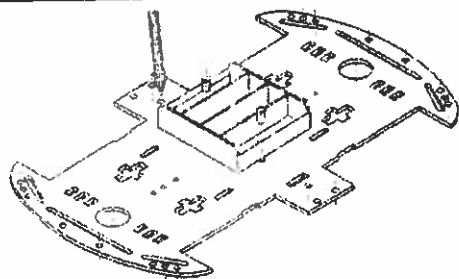
2、安装码盘，将电机固定在小车上，用 M3 螺丝拧紧。
(注意：电机引线铜片朝向外，即轮子一端。)



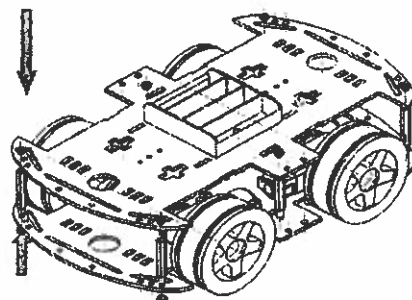
3、把电机固定到小车底盘上，插入螺丝，拧紧螺母。



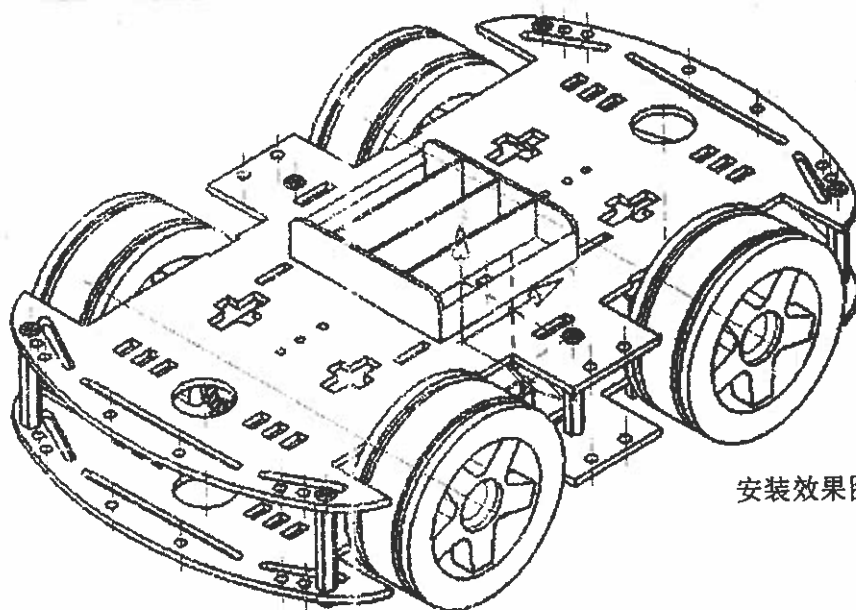
4、将轮胎固定在电机轴上。
(注意：轮子应该对准电机轴，电机轴与轮子插孔为椭圆形。)



5、用螺丝螺母将电池盒固定在底盘上。



6、放入铜柱下面上螺母，上面上螺丝固定上下板，并插入轮子，安装完成。



安装效果图

恭喜您已经完成 ZK-4 的组装，当然还有其他的组装方法，还在等什么，赶快测试吧，祝您成功！

注意事项：请勿从高空跌落，底盘材料为有机玻璃；电机请勿使用超过 15V 的供电系统使用，容易造成电机烧毁。导线在条件允许下，请焊接在电机上，请务必注意焊接安全，若没有焊接工具也可以直接绑在电机的两端使用。
技术支持与服务：若遇到小车载装或使用上的问题请联系当地的经销商或网络经销商。